

Lecture 14

Real Backend: API, Layers, EF Core and Dependency Injection

Continuation of the layered Expense Tracker

Dependencies point inward; concrete choices live at the edge.

- Combine API, EF and layers into a coherent host.
- Use DI lifetimes deliberately.
- Pass cancellation tokens through I/O.

Domain: stable business data.

Application: services and interfaces.

Infrastructure: EF implementation.

Presentation: endpoints.

Project: registers concrete services.

Scoped: one request and one DbContext.

Singleton: one app-wide instance.

Transient: a new resolution.

Project is the composition root.

- Read the runnable CodeExamples project first.
- Run the Project and identify where each responsibility lives.
- Change one implementation detail without breaking the Application layer.
- Write a short explanation of the boundary you protected.

- Apply the same decision to the student's chosen domain.
- Keep a screenshot or short note for the evolution portfolio.
- Use the database-session model and constraints.
- Prepare one demonstration scenario for the final defense.